

EC571: Digital VLSI Circuit Design

Homework 3 – Fall 2026

Total: 100 points

Due: October 11, 2026 at 11:59 pm

General Instructions

Use this worksheet to complete the homework. Please print this worksheet, write your solutions on the worksheet, take a picture of the worksheet, and submit it through Gradescope. If you prefer, you can download the Homework 3 worksheet on your tablet, complete the worksheet on your tablet, and submit the completed worksheet through Gradescope.

Show all reasoning. A final numerical answer without intermediate steps will receive limited credit. For power questions, clearly distinguish **energy per transition**, **average dynamic power**, **short-circuit power**, and **leakage/static power**. For logic-style questions, clearly state which devices conduct for the relevant input cases.

You may use calculators, lecture notes, and lecture slides. You may use AI tools only for concept review. You may not use AI tools to produce final derivations, final numerical answers, final circuit interpretations, or final written explanations. At the end of your submission, include one sentence disclosing whether and how you used AI tools.

Question 1: CMOS Power Accounting in an Activity-Dependent Block [20 points]

A digital block contains three classes of nodes. The supply voltage is $0.9V$, and the clock frequency is $1.2GHz$.

Node class	Number of nodes	Capacitance per node	Activity factor
local logic outputs	20,000	3 fF	0.12
medium wires	4,000	18 fF	0.08
long/global wires	600	95 fF	0.03

Assume each listed capacitance is the total capacitance switched by that node. In addition, the block has short-circuit power equal to 8% of the total dynamic switching power under nominal input slopes.

The leakage current per minimum-equivalent transistor is $2.5nA$. The block contains 1.8×10^6 minimum-equivalent transistors.

- (a) **Dynamic power by node class.** Compute the dynamic power contributed by each node class. Which class dominates? Explain whether the answer is obvious from capacitance alone. [6 points]

- (b) **Total dynamic and short-circuit power.** Compute total dynamic switching power. Then compute short-circuit power. [4 points]

- (c) **Leakage power.** Compute total leakage power. Compare leakage power against dynamic switching power. [4 points]

- (d) **Design interpretation.** Suppose a designer reduces the activity of the long/global wires by 50% using clock gating or operand isolation, but this adds 5,000 local logic nodes with activity factor 0.05 and capacitance 3 fF. Does this change reduce total dynamic power? Show your reasoning. **[3 points]**

- (e) **Reasoning check.** A student says: “The local logic has the most nodes, so it must dominate power.” Explain why this statement may be wrong using your numerical results. **[3 points]**

Question 2: Short-Circuit Current, Input Slope, and Load Dependence [20 points]

Short-circuit current occurs when both pull-up and pull-down networks conduct during an input transition. The short-circuit current waveform depends on the input slope and the load capacitance.

For an inverter, assume the short-circuit current pulse can be approximated as triangular. The supply voltage is $1.0V$.

A simulation produces the following measurements:

Case	Input transition time	Load capacitance	Peak short-circuit current	Short-circuit pulse width
A	20 ps	20 fF	$90 \mu A$	18 ps
B	80 ps	20 fF	$210 \mu A$	70 ps
C	80 ps	200 fF	$95 \mu A$	55 ps

The node switches with activity factor 0.15 at 800 MHz.

- (a) **Short-circuit energy.** Compute the short-circuit energy per transition for cases A, B, and C. [5 points]

- (b) **Short-circuit power.** Compute average short-circuit power for cases A, B, and C. [4 points]

- (c) **Load dependence.** Compare cases B and C. They have the same input transition time but different load capacitance. Explain why the larger load can reduce the short-circuit current pulse even though it increases capacitive switching energy. [4 points]

- (d) **Dynamic energy comparison.** For cases A and C, compute the ordinary capacitive switching energy for a rising output transition. Compare the short-circuit energy with the capacitive switching energy in each case. [4 points]

- (e) **Design action.** Give two circuit-level ways to reduce short-circuit power. For each method, state one possible penalty. [3 points]

Question 3: Leakage and Subthreshold-Slope Reasoning [20 points]

Lecture 8 introduces subthreshold conduction and the slope factor. Assume a transistor has subthreshold slope 80 mV/decade. At $V_{GS} = 0V$, its leakage current is $5nA$.

- (a) **Leakage increase with gate bias.** Estimate the leakage current if V_{GS} increases from 0V to 80 mV, 160 mV, and 240 mV. [5 points]

- (b) **Effect of threshold shift.** A process corner effectively lowers the threshold voltage by 80 mV. Explain why this has approximately the same effect as increasing V_{GS} by 80 mV. Estimate the new leakage current. [5 points]

- (c) **Sleep-mode leakage.** A block contains 5×10^5 leaking devices. Estimate total leakage power at $V_{DD} = 0.9V$ for the nominal case and for the $-80mV$ threshold-shift case. [4 points]

- (d) **Performance/leakage tradeoff.** Why might a designer intentionally use lower-threshold devices on a critical path despite the leakage penalty? [3 points]

- (e) **Reasoning check.** A student says: “An 80 mV threshold shift is small, so leakage changes only slightly.” Use your result to respond. **[3 points]**

Question 4: Interconnect Capacitance, RC Delay, and When Wires Matter [20 points]

A gate drives a wire of length L . The wire has resistance per unit length $0.08\Omega/\mu m$ and capacitance per unit length $0.20fF/\mu m$. The driver has intrinsic gate delay $35ps$.

- (a) **Wire delay for three lengths.** Compute the wire delay for:

$$L = 500\mu m, \quad 2mm, \quad 5mm.$$

Use consistent units and clearly show your reasoning.

[6 points]

- (b) **Critical length.** Find the wire length at which the wire delay becomes equal to the intrinsic gate delay.

[4 points]

- (c) **Capacitance and dynamic power.** For each of the three lengths in part (a), compute the wire capacitance. Assuming $V_{DD} = 0.9V$, activity factor 0.1, and frequency 1 GHz, compute dynamic power due to the wire capacitance. [4 points]

- (d) **Local versus global interconnect.** Explain why local interconnect and global interconnect scale differently as technologies scale. Your answer should refer to the conceptual distinction between local interconnect scaling with technology and global interconnect scaling with die size. [3 points]

- (e) **Design action.** For the 5 mm wire, propose two fixes to reduce delay. Examples include repeater insertion, wider wire, using a higher metal layer, or architectural floorplanning. For each fix, state one tradeoff. **[3 points]**

Question 5: Static CMOS, Ratioed Logic, and Pseudo-NMOS Design Tradeoffs [20 points]

Consider implementing the logic function:

$$Y = \overline{A + B + C + D}$$

with two alternative styles:

Option 1: Static complementary CMOS 4-input NOR.

Option 2: Pseudo-NMOS NOR with one always-on PMOS pull-up load and four NMOS pull-down devices in parallel.

Assume $V_{DD} = 1.0V$. For the pseudo-NMOS gate, the effective pull-up resistance is $30k\Omega$. Each ON NMOS pull-down has resistance $10k\Omega$. The output load capacitance is $40fF$.

For the static CMOS NOR, assume the worst-case pull-up resistance is $40k\Omega$, and the worst-case pull-down resistance is $10k\Omega$.

- (a) **Logic levels for pseudo-NMOS.** Compute V_{OL} for the pseudo-NMOS gate when exactly one NMOS pull-down is ON. Then compute V_{OL} when all four NMOS pull-down devices are ON. Explain why V_{OL} is input-pattern dependent. [5 points]

- (b) **Static power in pseudo-NMOS.** Compute static power in the pseudo-NMOS gate for the one-pull-down-ON case and the four-pull-down-ON case. [4 points]

- (c) **Delay comparison.** Estimate the falling-output delay for both designs when one NMOS pull-down is ON. Estimate the rising-output delay for both designs. Clearly state what resistance you use for each case. [5 points]

- (d) **Design tradeoff.** Which implementation would you choose for a low-power design? Which might be attractive for reduced transistor count or input capacitance? Explain. [4 points]

- (e) **Beyond fan-in four.** Suppose the function becomes:

$$Y = \overline{A + B + C + D + E + F}.$$

Explain why directly implementing this as a single high-fan-in gate is unattractive, and propose a better decomposition. [2 points]

Question 6: Complete the Cadence Virtuoso Tutorial 1 posted on Blackboard.